

Adaptive Baseline Affect Architecture

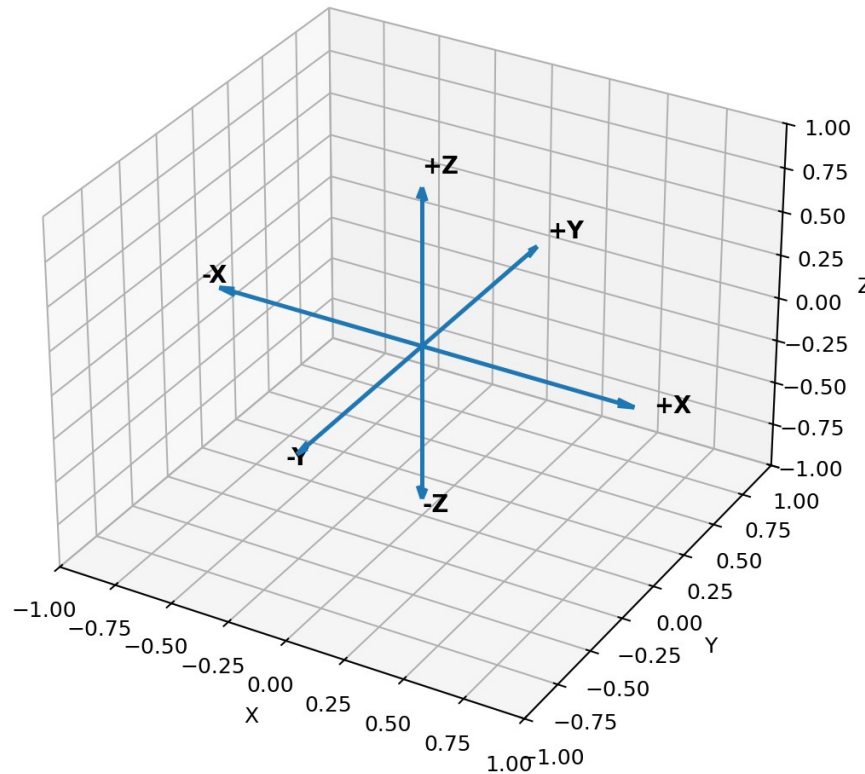
Theory • Master Detail Specification • Research Roadmap

Michael E Anderson

Manderson.tech82@gmail.com

www.github.com/CapAhabb

Six-Directional Computational State Space



Affective Intermediate Language (AIL) Emo-AI will use a compact, machine-readable shorthand for affective state, context, transitions, and memory significance. The goal is to preserve meaning while reducing repetitive natural-language token overhead, lowering inference cost and, where deployment scale makes it material, reducing associated compute and electricity use. Longhand explanations remain available for audit and validation.

Example: THR+.7 > FEAR.5 | TRUST-.2 | !.8 = threat escalation, recipient fear, reduced trust, high memory significance.

Concept specification for research and prototyping. Defined architecture is separated from hypotheses that require empirical validation.

Core Design Principle: Keep the computational architecture fixed while allowing the subject-relative reference frame to adapt.

Declarations & Intellectual Property Notice

This document presents the conceptual architecture, research direction, terminology, and proposed implementation framework for the Emo-AI Adaptive Baseline Affect Architecture.

Concept development and authorship credited to:

Michael E. Anderson

Manderson.tech82@gmail.com

www.github.com/CapAhabb

The ideas presented herein are developmental and exploratory. References to proposed mechanisms, terminology, architectures, and potential research contributions should not be interpreted as claims of validated scientific performance or confirmed novelty unless independently established through research, experimentation, and prior-art review.

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1. Executive Summary

Emo-AI proposes a unified affective representation for emotional detection, reasoning, persistence, and conveyance. The architecture uses six fixed computational directions (+X, -X, +Y, -Y, +Z, -Z), each bounded from 0.0 to 1.0, while allowing each subject to maintain a learned local baseline that functions as its own zero point.

Why it matters: A fixed zero forces every subject into the same reference frame. A movable local zero preserves a common computational language while allowing different subjects, contexts, and histories to occupy different resting states.

What We Want to Accomplish

- Create one affective intermediate language that can be used in both directions: observation → state and state → expression.
- Let the model help discover which recurring social-affective dimensions it can reliably perceive instead of hard-coding a complete emotional ontology in advance.
- Separate immediate state from slower learned disposition by allowing current state to decay toward a baseline while significant experience can gradually move the baseline itself.
- Preserve mixed or opposing states without forcing them to cancel into a single bipolar score.
- Test whether the same representation can later become a persistent self-state for the same base model without changing the core computational language.

2. Core Theory

The key architectural distinction is between a fixed global computational space and a dynamic subject-relative interpretation. Dimensionality remains constant; the local origin changes.

Layer	Fixed	Adaptive
Computational geometry	Six directional channels	No
Raw channel bounds	0.0–1.0	No
Subject baseline	Stored in same coordinate system	Yes
Local zero	Defined by baseline	Yes
Immediate state	Measured relative to baseline	Yes
Long-term disposition	Encoded by baseline migration	Yes

2.1 Six-Directional State Space

Six-Directional Computational State Space

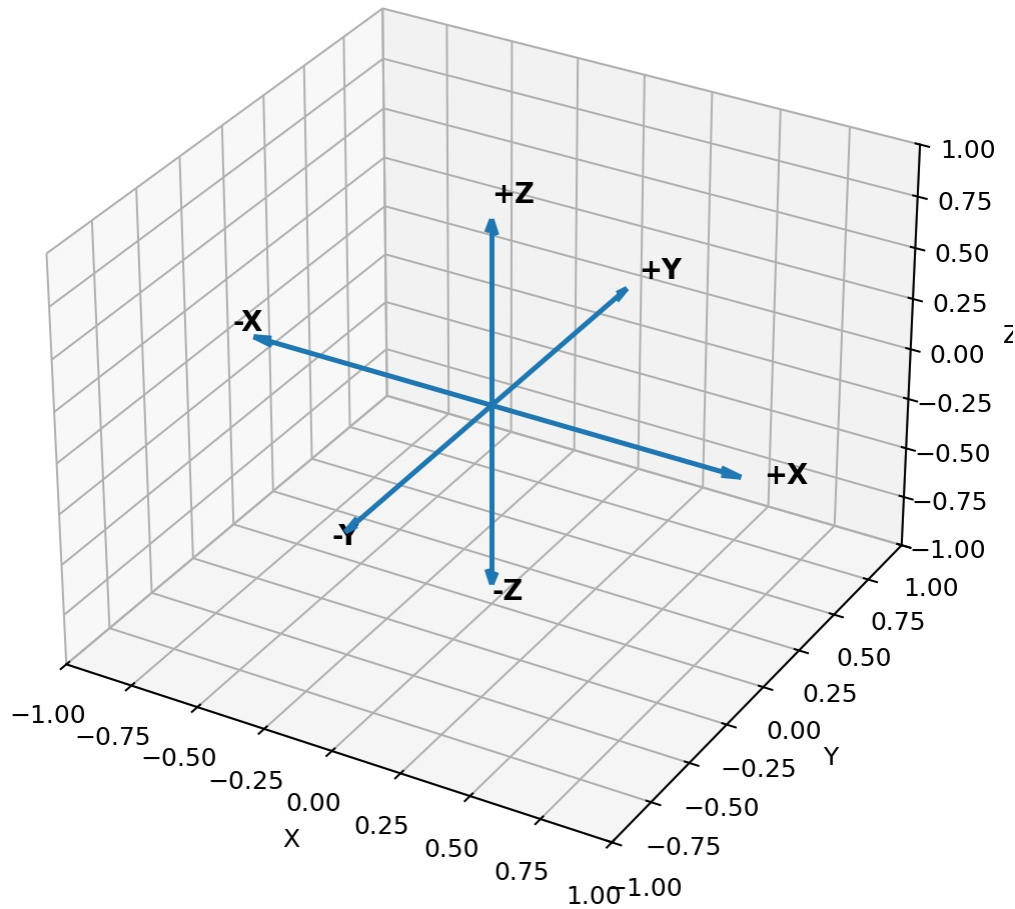


Figure 1. Fixed six-direction computational space. Semantic mappings should be validated rather than assumed prematurely.

Positive and negative directions are represented separately. This prevents mixed states from being collapsed by arithmetic cancellation. A subject can simultaneously carry activation in +X and -X if the observed interaction supports both.

3. Adaptive Local Zero

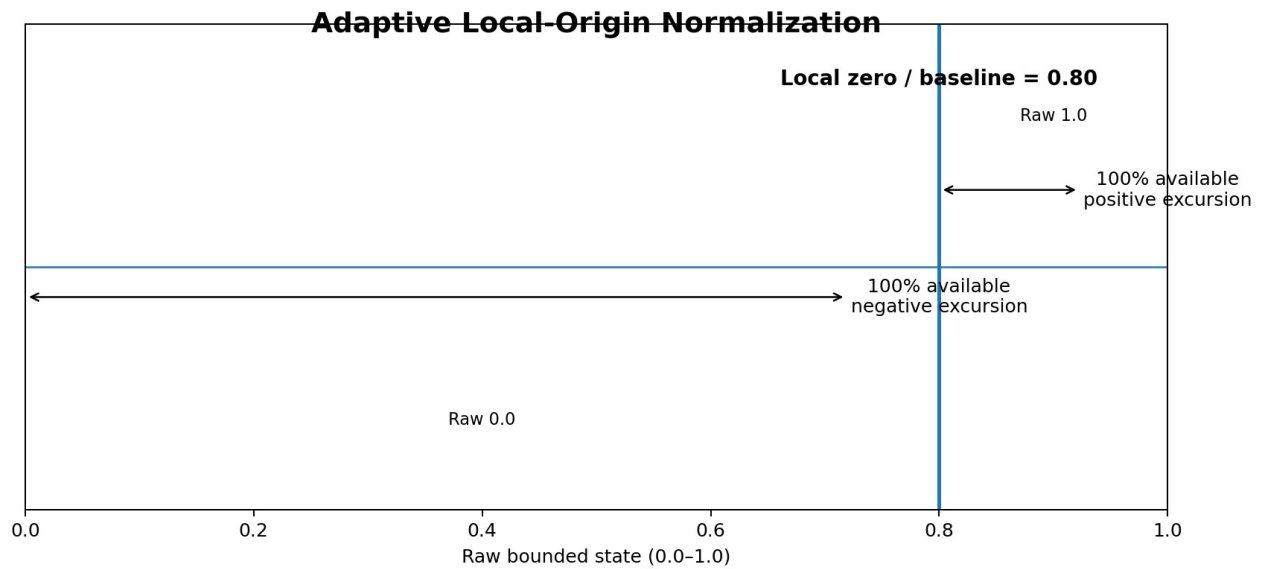


Figure 2. Example where raw 0.80 is interpreted as the subject's current local zero.

3.1 Raw State, Baseline, and Relative Expression

Every channel preserves three distinct quantities:

- Absolute state — the subject's global position in the bounded 0.0–1.0 coordinate system.
- Baseline — the subject's current resting reference point.
- Relative expression — the magnitude and direction of current displacement from that baseline.

Basic displacement: $D = S - B$

Directional normalization above baseline: $R+ = (S - B) / (1 - B)$

Directional normalization below baseline: $R- = (B - S) / B$

Interpretation: If $B = 0.80$, then raw 0.80 is local zero. Movement toward 1.0 and movement toward 0.0 can each still be normalized to a full 0–1 relative excursion without changing the underlying raw bounds.

4. Temporal Dynamics: State vs. Disposition

The architecture uses at least two timescales. Immediate affective activation should respond quickly and normally decay toward baseline. The baseline should move much more slowly and only when experience is sufficiently intense, repeated, prolonged, or significant.

Component	Timescale	Primary Driver	Function
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Current state	Fast	Immediate events	Momentary affect / reaction
EmoEcho	Medium	Persistence + reinforcement	Keeps affect active after the triggering moment
Baseline	Slow	Significant or repeated experience	Learned disposition / adaptation
Durable memory	Long	Memory-gate threshold	Preserves why meaningful baseline changes occurred

4.1 Baseline Migration Example

A trust-like channel may begin with $B = 0.80$. A severe betrayal moves current state to 0.15. Acute activation later decays, but if the event is sufficiently significant the baseline may settle at 0.55. The subject can again be at relative zero while remaining globally different from its earlier state.

5. System Architecture

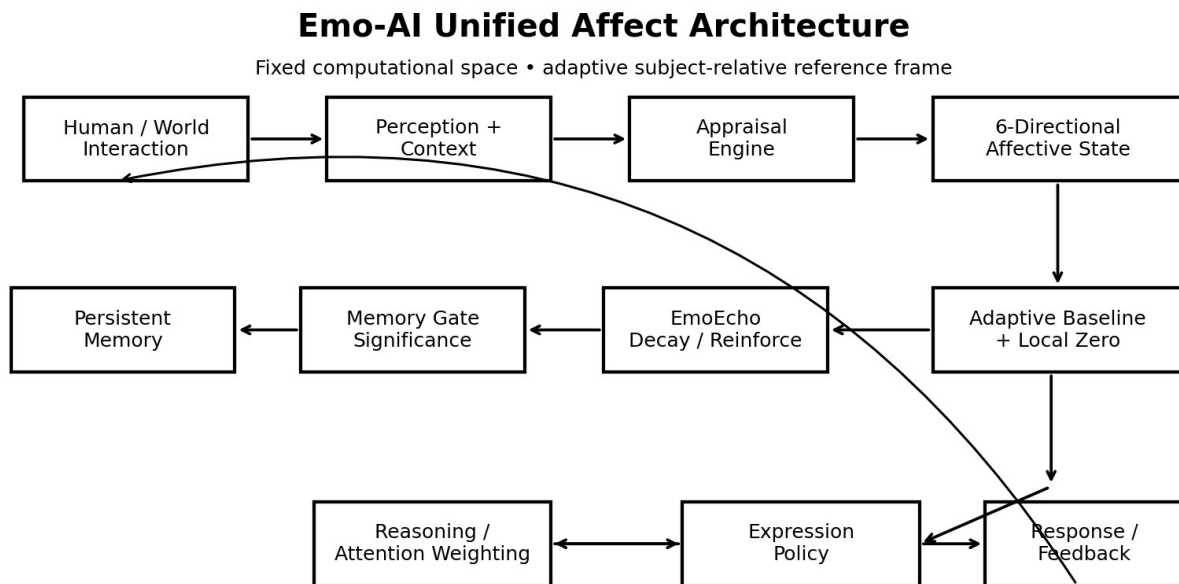


Figure 3. Unified pipeline from perception through appraisal, persistent state, memory, reasoning, expression, and feedback.

Perception + Context

Extract the observable interaction, participants, relationship context, relevant history, environment, and sequence of events.

Appraisal Engine

Determine why the event matters: benefit/harm, controllability, agency, intention, uncertainty, social meaning, threat, care, and likely consequences.

Six-Directional State

Encode the inferred affective state in the fixed six-direction bounded representation.

Adaptive Baseline

Interpret current activation relative to the subject’s learned local zero while preserving the global coordinates.

EmoEcho

Maintain short-to-medium affective persistence with decay and reinforcement rather than dropping affect immediately after inference.

Memory Gate

Promote only sufficiently significant interactions into durable memory. Preserve causes and evidence, not merely final coordinates.

Reasoning / Attention

Allow relevant affective state to influence prioritization, interpretation, and response selection without replacing task reasoning.

Expression Policy

Translate a desired state and context into language, tone, pacing, behavioral strategy, and other available output channels.

Feedback

Observe the recipient response and feed it back into the next appraisal cycle.

6. Observer-First Research Model

The first research system does not treat the AI as the emotional subject. It acts as an observer of human interaction. This keeps the project capability-constrained: the model first demonstrates what distinctions it can perceive, reason about, and score consistently.

6.1 Observation Record

Field	Question / Purpose
Initial state	What does the conveyor appear to be experiencing before the key interaction?
Context	Relationship, history, environment, stakes, goals, power, uncertainty.
Appraisal	What does the event mean to the conveyor and why?

Conveyance	How is the feeling established: words, tone, pacing, silence, body language, actions, mismatch?
Intent	Protect, warn, intimidate, reassure, repair, withdraw, seek help, gain compliance, etc.
Recipient interpretation	What does the recipient appear to believe the conveyor means or feels?
Recipient reaction	Immediate emotional and behavioral response.
Recipient impact	Longer effect on trust, safety, closeness, shame, confusion, fear, or cooperation.
Feedback to conveyor	How does the recipient's response change the original person's state?
Duration / decay	Onset, peak, persistence, reinforcement, recurrence, recovery.
State transition	What does the state become over time?
Memory significance	Should this interaction influence future behavior or baseline?
Confidence	How reliable is the inference? Preserve uncertainty separately from affect.

7. Construct Discovery and Psychological Grounding

Psychology should be used as a reference map and validation source, not as a rigid ontology imposed before testing. The observer model should identify recurring distinctions; those candidate constructs can then be compared with established work in appraisal, dimensional affect, interpersonal regulation, trust, attachment, empathy, threat perception, and social cognition.

Research discipline: The model may propose candidate dimensions, but no construct becomes part of the architecture until it survives consistency, predictive, transfer, and human-review tests.

8. Unified Detection and Conveyance Language

The core architectural claim is bidirectionality. Recognition and expression should not use separate emotional vocabularies.

Mode	Transformation	Purpose
Detection	interaction → affective state	Infer subject-relative state from evidence

Conveyance	affective state + context → expression	Generate behavior that communicates the intended state
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8.1 Round-Trip Test

- Observer Model analyzes an interaction and produces state vector V.
- Generation Model receives V plus minimal context and produces new behavior.
- An independent observer re-analyzes the generated behavior.
- Compare the reconstructed state to V.
- High reconstruction fidelity would support the claim that the representation functions as a common affective intermediate language.

8.2 Affective Intermediate Language (AIL)

Emo-AI will define a compact domain-specific intermediate language so recurrent affective states do not have to be re-described in verbose natural language. AIL is intended to be human-auditable, machine-parseable, versioned, and reversible into explanatory longhand. The shorthand is an encoding layer over the six-direction state model, not a replacement for it.

Efficiency Objective

AIL is designed to reduce token count for state descriptions, event transitions, memory entries, dataset annotations, and inter-module messages. Fewer tokens can reduce model computation for equivalent exchanges. Actual cost, latency, and electricity savings must be benchmarked on the selected model, tokenizer, hardware, context length, and deployment pattern.

8.3 Core Grammar

Token	Meaning	Example
S	Subject / actor	S1>S2
B	Learned baseline / local zero	B[x+.80 ...]
C	Current absolute state	C[x+.91 ...]
D	Baseline-relative expression	D[x+.55 y+.31]
T	Persistence / decay	T:.40
M / !	Memory significance	!.82
@	Context binding	@conflict
> / <>	Transition / mutual effect	THR+.7 > FEAR.5

Suggested operators: + increase/positive activation, - decrease/negative activation, ~ decay toward baseline, > causal or temporal transition, <> reciprocal influence, ! high significance, ? uncertainty, and @ context binding.

8.4 Longhand vs. Shorthand Examples

#	Longhand	AIL shorthand
1	The actor interprets the interaction as a strong threat. The recipient becomes moderately fearful, trust falls slightly, and the event is likely to be retained as highly significant.	THR+.70 > FEAR.50 TRUST-.20 !.80
2	Subject 1 is speaking to Subject 2 during conflict. The likely intent is protective, but the expression is highly activated and the recipient interprets it as threatening.	S1>S2 @conflict INT:protect ACT+.72 RX:THR.64
3	Activation rises above the learned baseline, persists briefly, then decays to local zero without moving the long-term baseline.	D[y+.62] T:.35 ~B DB:0

The longhand form remains the audit and explanation layer. AIL is the transport and state language. A compliant implementation should be able to expand AIL into human-readable rationale with the supporting evidence and confidence that produced the code.

8.5 Three-Layer Language Design

Layer	Contents	Purpose
Numeric core	B, C, D, T, M, confidence	State, baseline, relative expression, persistence, significance, and certainty.
Operators	+ - ~ > < > ! ? @	Compact direction, transition, significance, uncertainty, and context.
Semantic aliases	THR, CARE, TRUST, FEAR, ANGER, etc.	Readable aliases layered over validated constructs without changing the numeric core.

8.6 Versioning and Reversibility Requirements

- Every AIL record must declare or inherit a schema version so shorthand meanings cannot silently drift as the research ontology evolves.
- Numeric state must remain recoverable independently of semantic aliases. Renaming a construct must not invalidate the underlying state history.
- AIL compression must not remove evidence, confidence, actor/recipient identity, time, or baseline information when those fields are required for interpretation.
- Any claimed token, cost, latency, or electricity reduction must be measured against an equivalent longhand representation using the same model and task.

9. Preliminary Data Model

The implementation should preserve raw, baseline, relative, evidence, and confidence values separately. A simplified state record follows:

```
AffectiveState {
  subject_id
  timestamp
  context_id
  current_state: { +x, -x, +y, -y, +z, -z }
  baseline:    { +x, -x, +y, -y, +z, -z }
  relative_state:{ +x, -x, +y, -y, +z, -z }
  confidence
  supporting_evidence[]
  alternatives[]
  memory_significance
}
```

10. Research & Development Roadmap

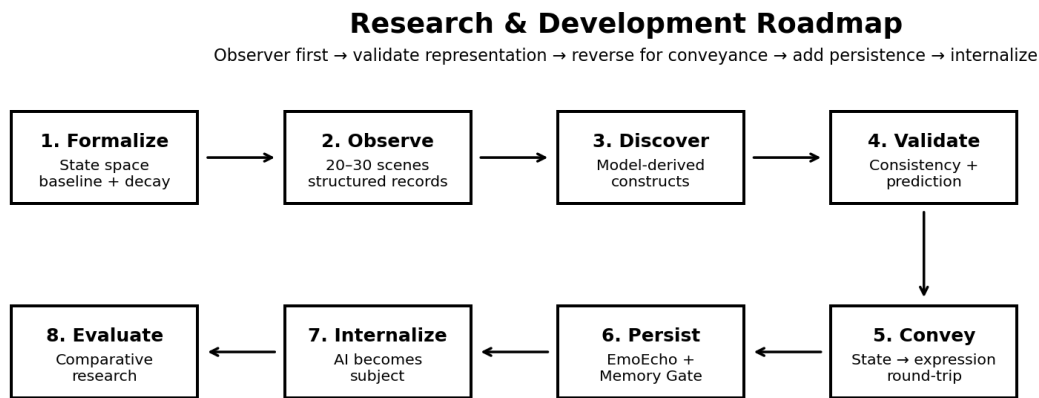


Figure 4. Staged development from formalization to self-state integration.

Phase 1 — Formalize Core Representation

Lock the state schema, baseline rules, directional normalization, confidence representation, decay terminology, and versioned data format.

Phase 2 — Observer Dataset

Start with approximately 20–30 carefully chosen scenes. Include clear and ambiguous examples of care, threat, frustration, fear, trust, betrayal, reassurance, sarcasm, misunderstanding, and repair.

Phase 3 — Construct Discovery

Let the same base model identify recurring distinctions and propose mappings for the six directions. Preserve model-human disagreement as research data.

Phase 4 — Validation

Test repeatability, calibration, recipient-reaction prediction, transition prediction, mixed-state handling, cross-context transfer, and baseline stability.

Phase 5 — Conveyance

Reverse the mapping. Generate behavior from state coordinates and perform perception → representation → generation → perception round-trip tests.

Phase 6 — EmoEcho + Memory Gate

Add persistence, decay, reinforcement, significance thresholds, and causal memory records.

Phase 7 — AI as Subject

Apply the same representation as a persistent model-relative state. Keep the same base model where practical to minimize translation between observer and subject representations.

Phase 8 — Comparative Research

Compare against discrete emotion labels and conventional dimensional baselines. Measure whether adaptive local-origin representation improves continuity, transfer, prediction, and expression fidelity.

11. Validation Framework

- Test/retest consistency
- Human inter-rater agreement
- Confidence calibration
- Recipient-response prediction
- State-transition prediction
- Duration/decay estimation
- Mixed-state preservation
- Cross-context generalization
- Cross-subject generalization
- Baseline stability and migration validity
- False threat/care inference rate
- Round-trip perception-expression fidelity

11.1 Comparative Baselines

- Model A — discrete categorical emotion labels.
- Model B — conventional fixed dimensional representation.
- Model C — Emo-AI adaptive local-origin six-direction representation.

The key research question is whether Model C provides measurable gains rather than merely a more intuitive visualization.

12. Engineering Constraints and Failure Modes

Risk	Control
Baseline drift	If the baseline moves too easily, temporary events become permanent personality changes. Require slow updates and strong evidence.
Saturation	Baselines near 0 or 1 create asymmetric raw room. Preserve global state but use directional normalization for relative expression.
Ontology overfitting	Do not force every interaction into predetermined emotional categories. Allow uncertainty and alternative interpretations.
False mind-reading	Emotional states are inferred, not observed facts. Store supporting evidence, alternatives, and confidence.

Dimensional creep	Add dimensions only when recurring, behaviorally meaningful information cannot be represented without loss.
Circular validation	Do not use the same model output as both prediction and ground truth. Use human review and independent evaluation paths.
Scripted-data bias	Movies are useful early because context is dense and repeatable, but real-world interaction data is required before strong generalization claims.

13. Research Contribution to Test

The project should not claim novelty simply for using numbers, multidimensional affect, emotional decay, or personalization. Those ideas have substantial precedent. The candidate contribution is the integration and geometry:

Candidate contribution: A fixed, bounded six-direction affective representation with dynamically learned subject-relative local origins, preserving global state while normalizing expression around adaptive baselines, and using the same representation bidirectionally for perception and conveyance.

A second candidate contribution is using observer-derived affective constructs learned by a model as the later persistent self-state representation of the same underlying model. Both require formal prior-art review and experimental validation before being described as novel.

14. Minimum Viable Research Prototype

Input

A short interpersonal scene with enough surrounding context to infer sequence and relationship.

Processing

- Identify participants and context.
- Estimate current and baseline six-direction state.
- Explain appraisal, expression, intent, recipient interpretation, recipient impact, and feedback.
- Estimate duration, transition, memory significance, confidence, and alternative explanations.

Storage

Versioned JSON records containing state, baseline, relative state, evidence, confidence, and human corrections.

Output

A state trajectory plus a human-readable rationale. No AI self-state is required in the first prototype.

15. Recommended First Sprint

- Freeze Version 0.1 terminology: six directions, raw state, baseline, local zero, relative expression, EmoEcho, Memory Gate, confidence.
- Define a machine-readable observation schema.
- Select 20–30 initial interactions spanning clear and ambiguous cases.
- Run repeated observer passes with the same model and compare consistency.
- Record human corrections and model-human disagreements rather than smoothing them away.
- Only after the first corpus is reviewed, decide whether X/Y/Z semantics are stable enough to name.
- Do not add self-state or persistent model behavior until perception and representation pass basic validation.

16. Design Statement

Emo-AI keeps the computational architecture fixed while allowing the reference frame to learn. Immediate state moves quickly. EmoEcho preserves meaningful activation. Significant experience can move the baseline. Memory preserves why. The same affective language is used to observe, reason, remember, and eventually convey. AIL provides the compact shared syntax used to carry that state between observation, memory, reasoning, and expression.

Working architecture: Fixed dimensionality. Dynamic local zero. Preserved global coordinates. Bidirectional affect language.